



Information Technology Institute (ITI)
Analog IC Design Track
Summer Camp 2026
Lab 2
Common Source Amplifier



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PART 1: Sizing Chart

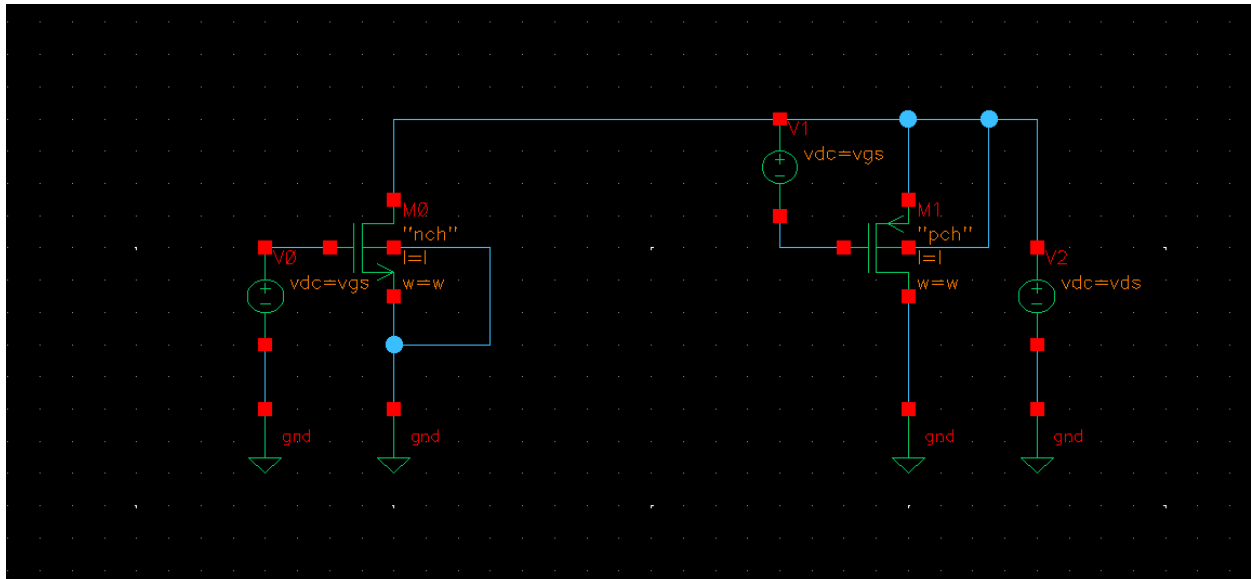


Figure 1: schematic

- Design parameters:
 - $V_{dd} = 1.8 \text{ v}$
 - $I_d = 86 \text{ uA (Random)}$
 - $|A_v| = 5 = g_m \times R_D \text{ (Random)}$
 - $V_{RD} = 0.9 \text{ v}$
 - $L = 2 \text{ um}$
 - $R_D = \frac{0.9}{86 \text{ u}} = 10.465 \text{ k}$
 - $|A_v| = 5 = 2 \frac{V_{RD}}{V_*} \rightarrow V_* = 360 \text{ mv}$



- 1) Plot V^* and V_{ov} overlaid vs V_{GS} . Make sure the y-axis of both curves has the same range.

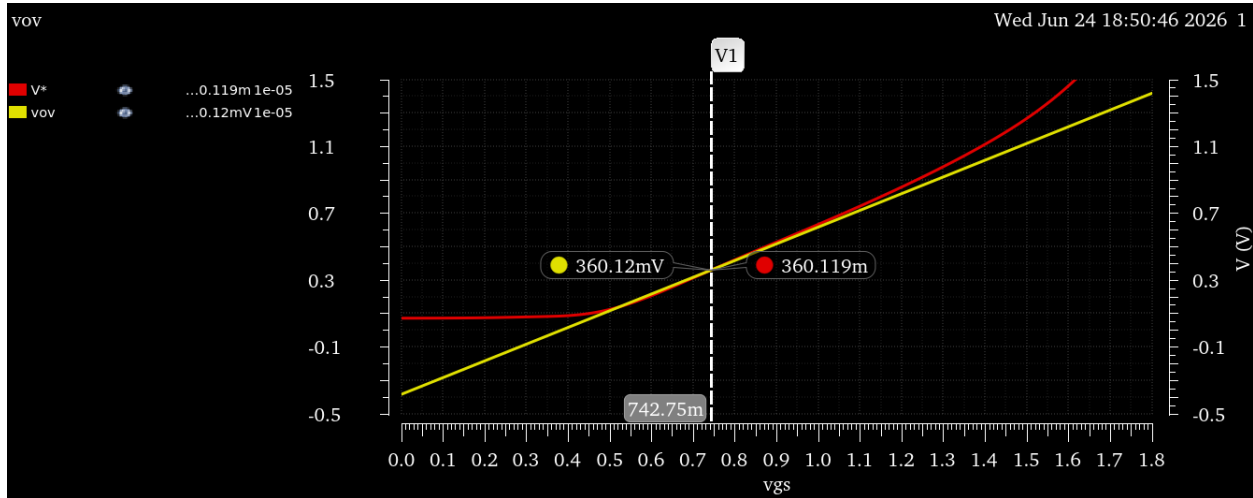


Figure 2 : V^* and V_{ov} vs V_{GS}

→ $V_{GS} = 742.75$ mV

- 2) Plot I_D , g_m , and g_{ds} vs V_{GS} . Find their values at V_{GSQ} . Let's name these values I_{DX} , g_{mX} , and g_{dsX} .

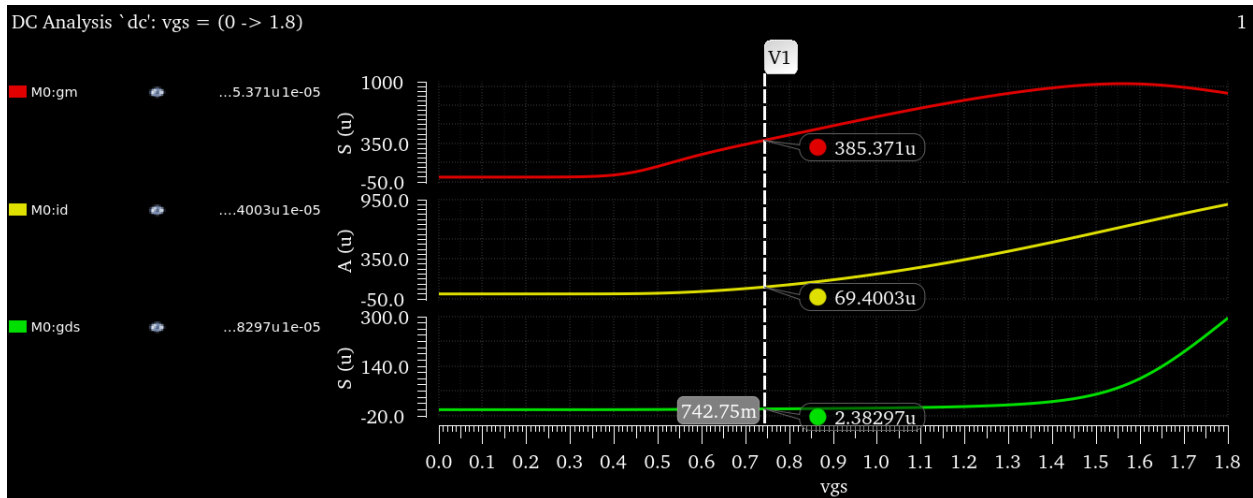


Figure 3 : I_D & g_m & g_{ds} vs V_{GS}

→ $I_{DX} = 69.4003$ uA @ $W = 10$ um



$$I_d \propto W \rightarrow W(\text{for } I_d = 86\mu A) = 10 \times \frac{86}{69.4003} = 12.392 \mu m$$

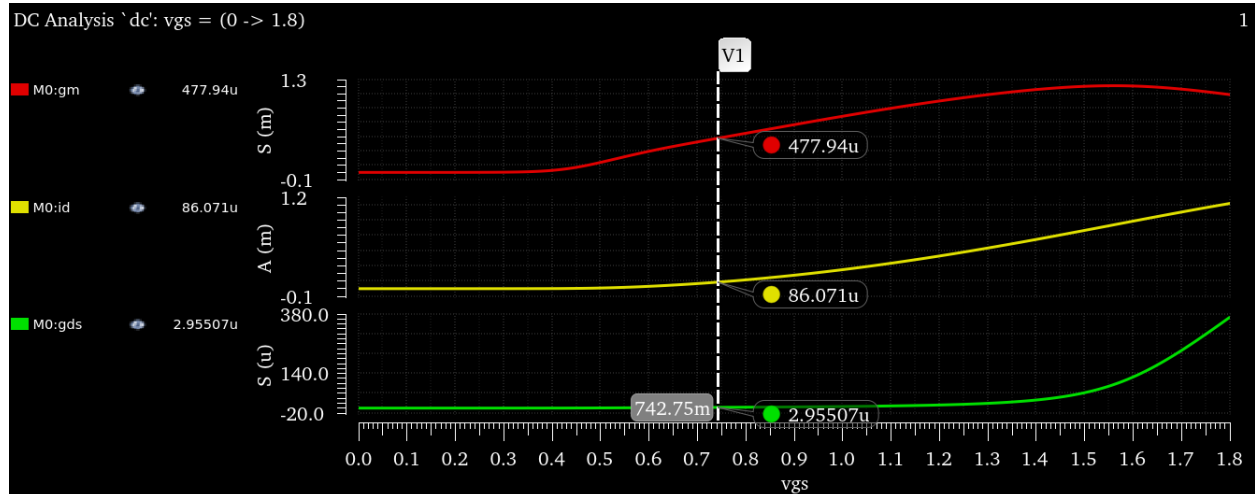


Figure 4: I_d & g_m & g_{ds} vs V_{gs} @ $W = 12.392 \mu m$

➔ Finally, nmos parameters:

- $W = 12.392 \mu m$
- $r_o = 1/2.95507 \mu m = 338.401 \text{ k ohm}$
- $g_m = 477.94 \mu S$
- $V_{gs} = 742.75 \text{ mV}$ (Biasing)
- $R_D = 10.465 \text{ k ohm}$



PART 2: CS Amplifier

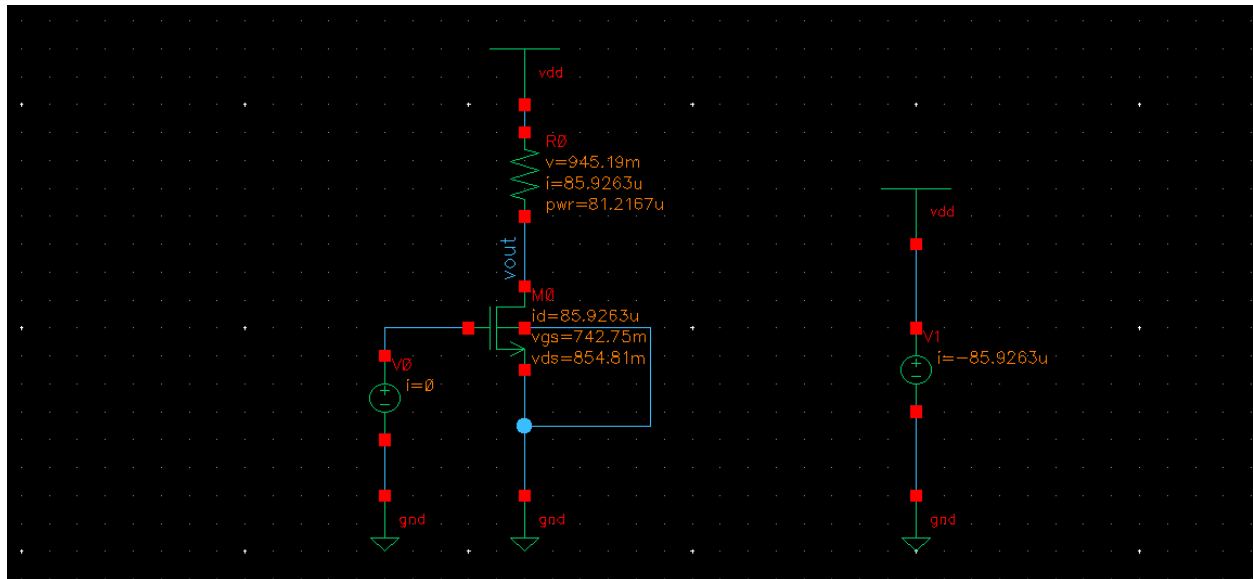


Figure 5: common source

1. OP and AC Analysis

1) Simulate the DC OP. Report a snapshot for the key operating point (OP) parameters.

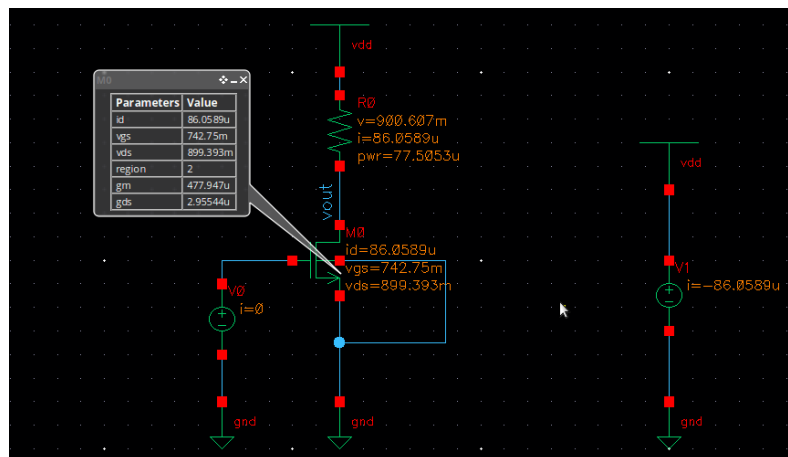


Figure 6: DC_OP



PARAMETER	CHART-BASED DESIGN(PART1)	CADENCE SIMULATION(PART2)
ID	86uA	86.0589uA
VGS	742.75 uv	742.75 uv
VDS	900 mv	899.393 mv
REGION	Saturation	Saturation
GM	477.94 uS	477.947 uS
GDS	2.95507um	2.95544 uS

2) Compare r_o and R_D .

→ $r_o = 338.4 \text{ k ohm}$ & $R_D = 10.465 \text{ k ohm}$ → $R_D // r_o \approx R_D$

Because; $R_D \ll r_o$

3) Do you expect the error to remain the same if we use min L?

→ No, the error is not expected to remain the same; it will significantly increase. The transistor's output resistance r_o is directly proportional to the channel length L due to the Channel Length Modulation (CLM) effect. If the minimum channel length ($\min L$) is used, the CLM effect becomes highly pronounced, causing r_o to drop substantially. As r_o decreases and becomes comparable in magnitude to R_D , the approximation $r_o || R_D \approx R_D$ is no longer valid. Consequently, ignoring r_o under minimum L conditions would yield highly inaccurate analytical results and a significant discrepancy between calculated and simulated gain.

4) Calculate the intrinsic gain of the transistor.

$$|A_v| = g_m \times r_o = 477.94 \times 10^{-6} \times 338.4 \times 10^3 = 161.734$$



- 5) Calculate the amplifier gain analytically. What is the relation ($\ll, \gg$) between the amplifier gain and the intrinsic gain?

$$|A_v|_{amp} = 477.947 \times 10^{-6} \times 10.465 \times 10^3 = 5$$

$$|A_v|_{intrinsic} = 161.734$$

intrinsic Gain $\gg$ amplifier Gain (Expected)

- 6) Create a new simulation configuration and run AC analysis (from 1Hz to 1GHz). Report the gain vs frequency. Annotate the DC gain and make sure it meets the spec.

➔ Before modification (using $R_D = 10.465 \text{ k ohm}$)

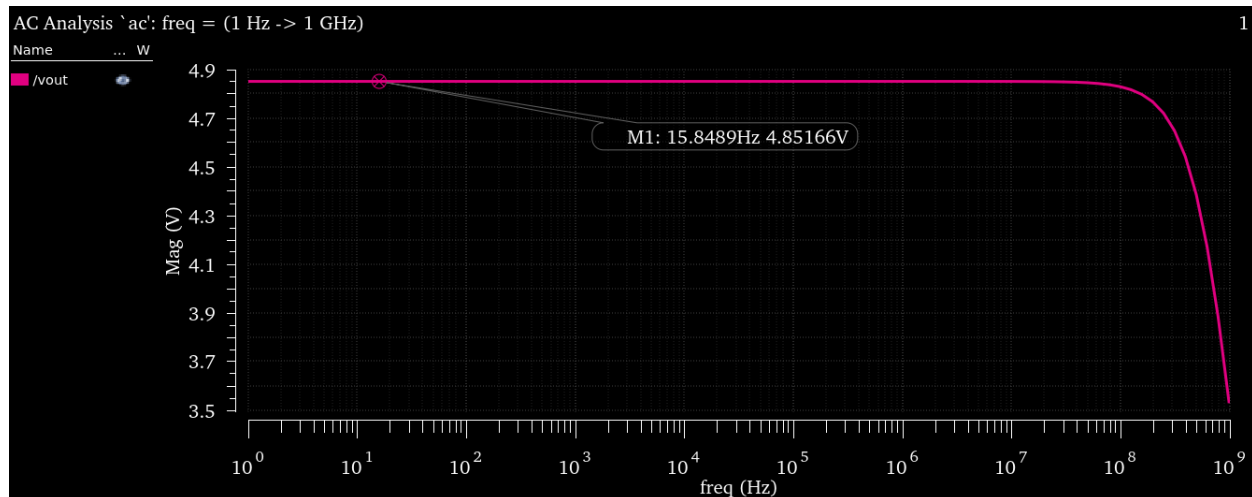


Figure 7: gain not equal 5

- (idea) Sweep R_D for reach gain = 5

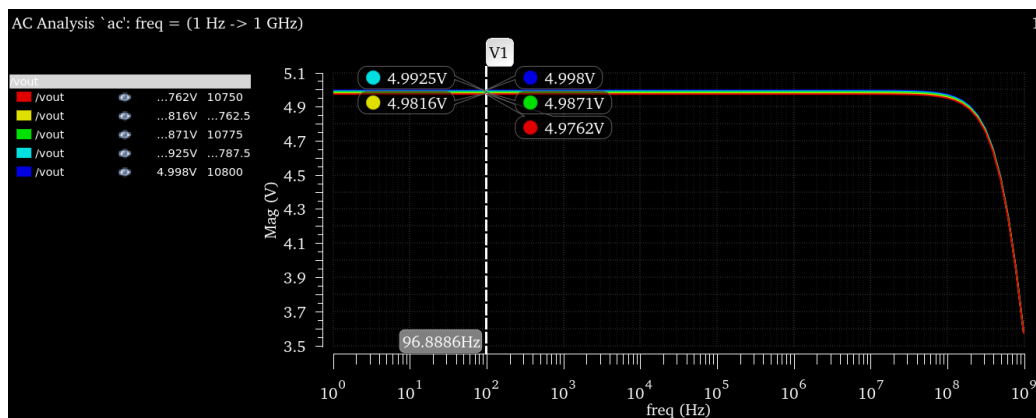


Figure 8: R_D sweep



→ after modification ($R_D = 10.81 \text{ k}\Omega$)

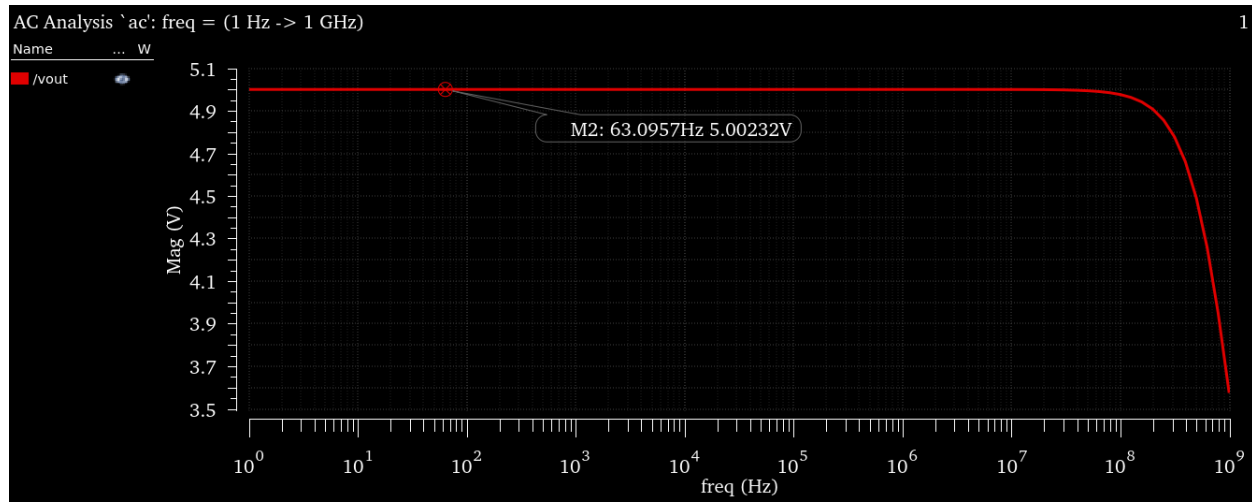


Figure 9: $R_D = 10.81 \text{ k}\Omega \rightarrow \text{gain } 5$

2. Gain Non-Linearity

1) Report V_{OUT} vs V_{IN} . Is the relation linear? Why?

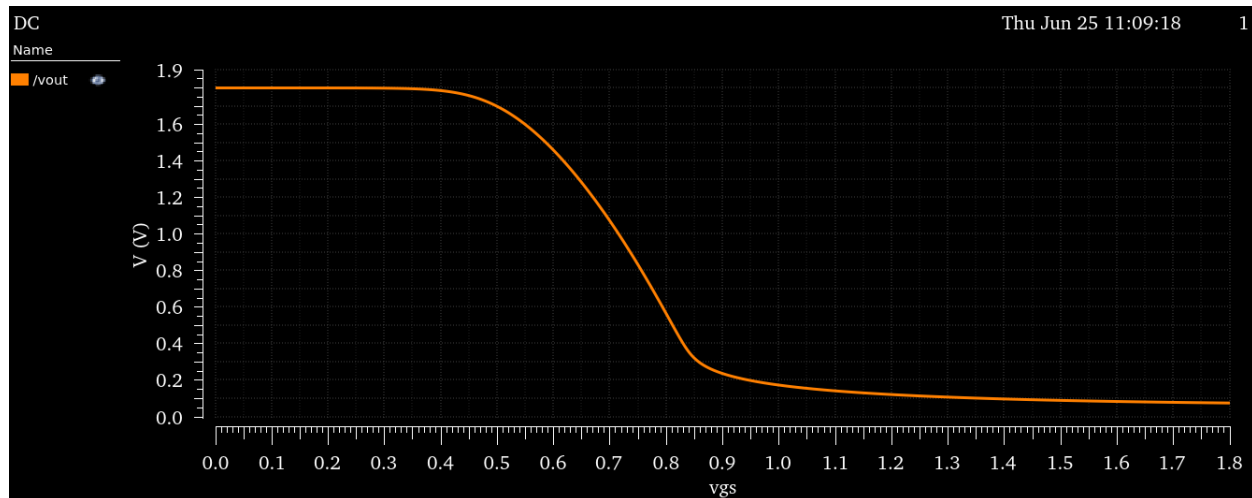


Figure 10: V_{out} vs V_{gs}

→ No, the relationship between V_{OUT} and V_{IN} is not linear.

The DC sweep illustrates the large-signal behavior of the amplifier. The non-linearity occurs because the MOSFET transitions through different operating



regions as the input voltage (V_{IN}) sweeps from 0 to V_{DD} . The governing equations for the drain current (I_D) change significantly in each region:

- **Cut-off Region:** When $V_{IN} < V_{TH}$, the transistor is OFF. Since $I_D = 0$, there is no voltage drop across the load resistor R_D , and the output remains constant at maximum: $V_{OUT} = V_{DD}$.
- **Saturation Region:** As V_{IN} increases beyond V_{TH} , the transistor turns ON and enters saturation. In this region, I_D exhibits a non-linear (square-law) dependence on V_{IN} . Since $V_{OUT} = V_{DD} - I_D R_D$, the output voltage decreases in a non-linear fashion. This steep, non-linear slope is where amplification occurs.
- **Triode Region:** As V_{IN} continues to increase, V_{OUT} drops significantly until $V_{OUT} < V_{IN} - V_{TH}$, pushing the transistor into the triode region. The I_D equation changes again, causing the V_{OUT} curve to gradually flatten out at a low voltage level.
- **Note:**
Linear amplification in a Common-Source amplifier is only a valid approximation for very small input variations (Small-Signal) around a specific, properly biased DC operating point, not for a full-scale voltage sweep.

2) Calculate the derivative of V_{OUT} using calculator.

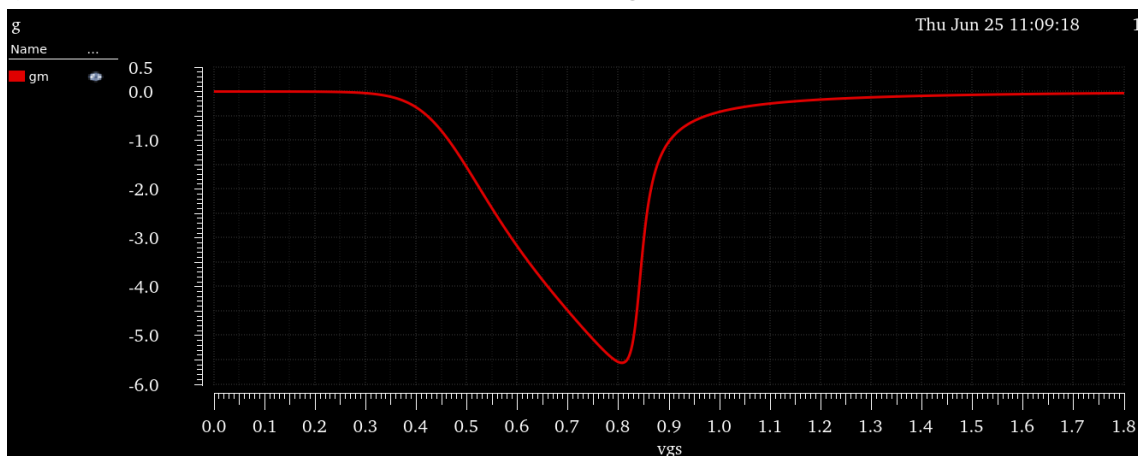


Figure 11: g_m vs v_{gs}



→ **Is the gain linear (independent of the input)?** No, the small-signal gain is **not linear**; it is highly dependent on the input voltage V_{in} . If the gain were independent of the input, the derivative plot would be a flat, constant horizontal line. Instead, the graph shows a curve that varies drastically with V_{in} .

→ **Why?** The derivative of the transfer characteristic curve represents the small-signal voltage gain A_v . For a resistively loaded Common-Source amplifier, the gain is approximately:

$$A_v = -g_m(r_o // R_D) \approx -g_m R_D$$

Since R_D is constant, the variation in gain is driven by the transconductance g_m , which is a strong function of the operating point V_{GS} or V_{in} . The plotted curve can be explained through the transistor's operating regions:

1. **Cut-off Region (Low V_{in}):** When $V_{in} < V_{th}$, the transistor is OFF. The transconductance g_m is zero, resulting in zero gain (the curve is at 0 on the y-axis).
2. **Saturation Region (Moderate V_{in}):** As V_{in} exceeds V_{th} , the transistor enters saturation. The value of g_m increases, causing the magnitude of the negative gain to increase significantly. The peak magnitude of the gain (the lowest point on the negative curve) occurs in this region, which is the optimal biasing point for linear amplification.
3. **Triode Region (High V_{in}):** As V_{in} continues to increase, the voltage drop across R_D increases, pushing V_{out} lower until the transistor enters the triode (linear) region. In the triode region, g_m drops rapidly and the transistor's output resistance r_o collapses. Consequently, the magnitude of the amplifier's gain sharply decreases back towards zero.

Conclusion: The gain is extremely sensitive to the DC bias point. To achieve relatively "linear" amplification, the input signal must be kept very small (Small-Signal) and centered exactly at the V_{in} that provides the peak gain in the saturation region.

3) Plot gm vs time

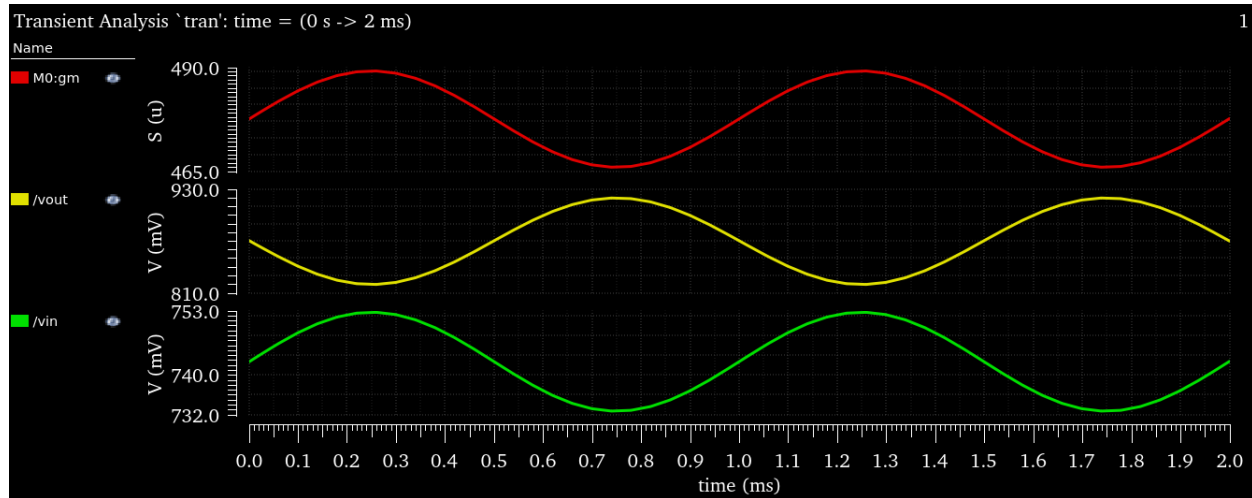


Figure 12: g_m , V_{out} & V_{in} vs time

4) Transient Variation of Transconductance g_m

Yes, the transconductance (g_m) visibly varies with the input signal over time, as shown in the transient analysis plot.

What this means: The transconductance of a MOSFET is fundamentally dependent on the instantaneous gate-source voltage $v_{gs}(t)$. While we often assume g_m is a constant value in small-signal analysis, this is only valid for extremely small perturbations around the DC bias point V_{GSQ} . When a transient stimulus with a 10 mV amplitude is applied, the instantaneous operating point shifts significantly back and forth along the non-linear $I_D - V_{GS}$ curve. Consequently, g_m fluctuates periodically. This indicates that the input signal amplitude is large enough to break the strict small-signal assumption, causing the amplifier's instantaneous gain $A_v \approx g_m(t) R_D$ to be time-varying rather than perfectly constant.

5) Amplifier Linearity Assessment

Strictly speaking, the amplifier is **not completely linear**. For an amplifier to be perfectly linear, its voltage gain must remain absolutely constant regardless of the input signal's instantaneous value. Since the gain is directly proportional to g_m , the observed variation of g_m with the input signal means the gain is dynamically changing during the positive and negative half-cycles of the input sine wave. This



dynamic change in gain introduces harmonic distortion, meaning the output sine wave is not a perfectly scaled replica of the input. The amplifier only approximates linear behavior if the input signal amplitude is kept infinitesimally small. The applied 10mV amplitude is large enough to expose the inherent non-linear large-signal characteristics of the MOSFET.